

JANANI ASOKAN

[Portfolio](#) | ☎ 6383495030 | ✉ jananiasokanse@gmail.com | [Github](#) | [Linkedin](#)

SUMMARY:

Passionate software engineer with hands-on experience in Software Testing, AI and DS. Proficient in Python and C++ with a focus on building scalable and Automation Test Scripts. Applying my technical skills to drive innovation in next-generation AI computing technologies

EXPERIENCE:

Software and System Test Intern, Signify, Bengaluru, India

July 2025 – May 2026

- Built an Integrated AI to automate the Firmware and Cloud API testing flow
- Built a scalable cloud-native API automation Test Script using Python and Pytest to validate RESTful APIs across dev, staging, and production, improving test coverage, reliability, and release quality.
- Validating data flow between MQTT brokers, cloud APIs, and mobile clients to ensure system reliability and synchronization.
- Implemented robust authentication and authorization handling (dynamic tokens, session management) and resilience mechanisms such as retries and response-time validation.
- Executed firmware test cases using MQTT protocol for real-time device communication ensuring reliable message validation

EDUCATION:

MTech Computer Science and Engineering | CGPA: 8.4 | Vellore Institute of Technology, Vellore

July 2021 – July 2026

SKILLS:

Languages: Python, C, C++

AI Tech: Machine Learning, Deep Learning, NLP, LLM, RAG, Generative AI, Agentic AI

Data Visualization: Exploratory Data Analysis (EDA), Power BI, Object Oriented Analysis and Design

Computer Science Fundamentals: Data Structures and Algorithms, Operating Systems, AWS

Web Development: HTML, CSS, JavaScript

Frameworks: TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, Matplotlib

Dev Tools: Docker, Kubernetes, CI/CD, DevOps, RESTful API, Wireshark, Packet Sender

Software: GitHub, VS Code, Google Colab, Postman API, Pytest, Firmware

Soft Skills: Clear technical communication, efficient debugging, consistent contributions to team goals

PROJECTS:

AI based Adaptive Traffic Signal Control System for Emergency Vehicle Prioritization using IoT | AWS IoT, YOLO V8, ML

- Engineered an IoT-based traffic control system using Reinforcement Learning to prioritize emergency vehicles, tested on 90-second video feeds.
- Facilitated automatic signal switching at 4-way intersections, cutting manual intervention and improving emergency response time by 30%.

Real-Time Handwritten Alphabet Recognition System and Object Detection | Deep Learning, Open CV

- Constructed a real-time handwritten recognition system with OpenCV and EMNIST, achieving 85% accuracy and identifying 20+ object classes.

IoT-Based Response Drone for Search and Rescue

- Developed an IoT-enabled drone system using Arduino for real-time disaster search and rescue operations
- Integrated an AI-driven **image processing** module to identify stranded individuals in disaster-affected areas with 95% accuracy.
- Conducted extensive field tests, simulations in flood conditions, tested the drone's performance 18,000 km away, demonstrating global operability. Enabled remote operation and deployment anywhere in the world

WebDigestAI - Efficient Content Digesting with AI-driven Insights | NLP, Python, Langchain, Langgraph

- Engineered an AI-driven article summarization tool that extracts content from diverse web sources, reducing time by 65%.
- Implemented real-time data scraping from 1000+ URLs using BeautifulSoup4 and WebBaseLoader
- Applied LangChain and RAG Model Methodology for advanced NLP. Integrated the Gemini API for 90% accurate responses.

AWARDS AND HONORS

- **PitchWar Hackat finalist** secured 3rd place by developing an IoT-enabled drone system with AI-driven image processing - Best Innovator Award
- **ISRO Hackathon finalist**, achieving **87.2% accuracy** in 30k moon image annotation using YOLO V8 [[Link](#)].
- **GEO Predict Hack** – ML powered web application for predicting geographical trends using real-time data analysis [[LINK](#)]
- **Research on Federated Learning** for Autonomous Vehicle Detection– Developed a decentralized AI model
- **CouchDB** - CouchDb(Remote Internship) [[LINK](#)]
- **ISTQB Certified Tester** - [[LINK](#)]